

ECON 470 Homework 5

Submission 1, Spring 2026

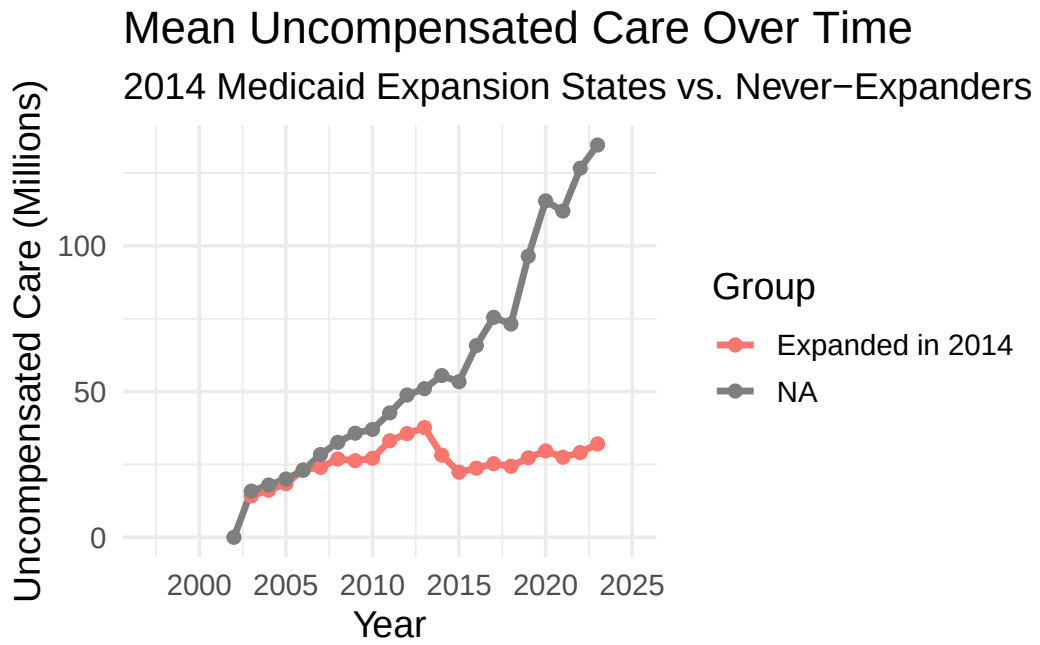
Srijon Sarkar

Problem 1

Table 1: Uncompensated Care (\$M), 2010–2018

Year	Mean (\$M)	Std. Dev. (\$M)
2010	30.26	72.26
2011	34.10	74.96
2012	37.54	86.63
2013	39.43	80.86
2014	36.63	88.53
2015	33.37	86.90
2016	45.24	403.12
2017	41.14	102.67
2018	39.10	100.40

Problem 2



Problem 3

Table 2: Difference-in-Differences: Mean Uncompensated Care (\$M)

Group	2012 (\$M)	2015 (\$M)	Difference (\$M)
Control	48.81	53.38	4.57
Treatment	35.58	22.35	-13.24

Problem 4

The adoption of Medicaid leads to a decrease in uncompensated care, as observed. Thus, we expect this effect as it encourages hospitals to take Medicare patients and get compensation or partial support from the Medicare program, instead of outsourcing debt to manage or counter their listed charges to a third-party bank.

Estimate ATEs

Problem 5

Table 3: DiD Estimates: Effect of Medicaid Expansion on Uncompensated Care (\$M)

Term	Estimate	Std. Error	t-Statistic	p-Value
Intercept	3.1725e+01	9.700e-01	3.2722e+01	2.27e-231 ***
Post 2014	4.8524e+01	1.576e+00	3.0786e+01	2.17e-205 ***
Ever Expanded	-5.9650e+00	1.251e+00	-4.7680e+00	1.87e-06 ***
Post x Ever Expanded (DiD)	-4.8266e+01	2.016e+00	-2.3943e+01	1.10e-125 ***

Problem 6

Table 4: TWFE Estimates: Effect of Medicaid Expansion on Uncompensated Care (\$M)

Term	Estimate	Std. Error	t-Statistic	p-Value
Treatment (DiD)	-5.2697e+01	1.0205e+01	-5.164e+00	7.96e-06 ***

We notice that both the estimators yield near about same magnitude of estimated effect around 48 to 52 millions of uncompensated care.

Problem 7

Table 5: OLS DiD: Effect of Medicaid Expansion on Uncompensated Care (\$M), All States

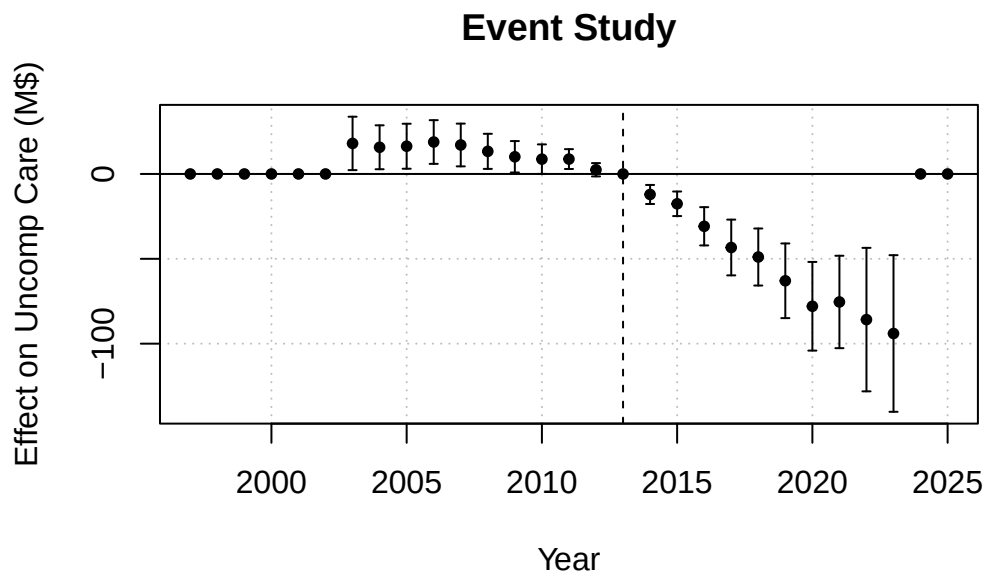
Term	Estimate	Std. Error	t-Statistic	p-Value
Intercept	31.725	1.352	23.463	4.94e-121 ***
Post 2014	48.524	2.198	22.074	1.98e-107 ***
Ever Expanded	-7.893	1.620	-4.872	1.11e-06 ***
Post x Ever Expanded (DiD)	-42.565	2.609	-16.314	1.16e-59 ***

Signif. codes: *** p<0.01, ** p<0.05, * p<0.10

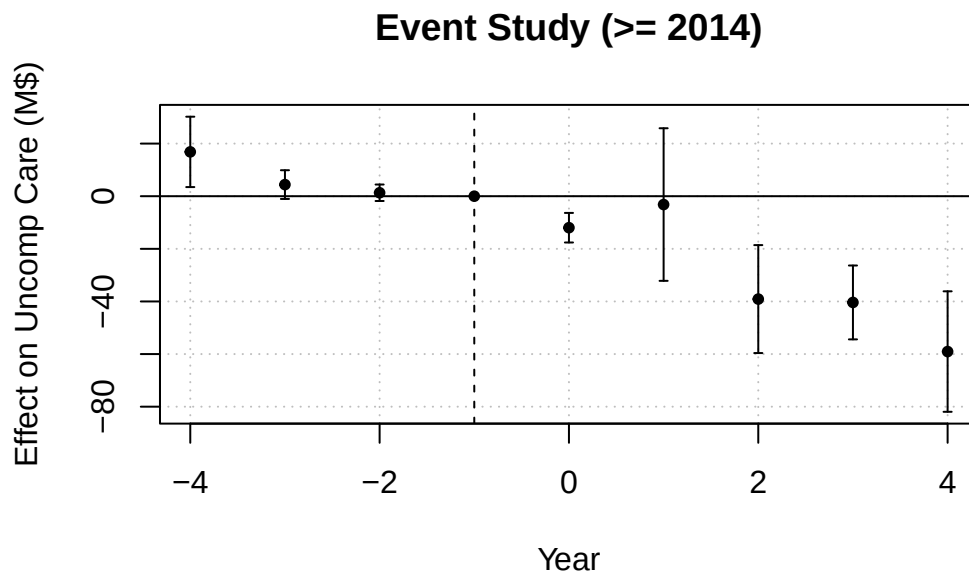
Table 6: TWFE: Effect of Medicaid Expansion on Uncompensated Care (\$M), All States

Term	Estimate	Std. Error	t-Statistic	p-Value
Treatment (DiD)	-40.038	9.812	-4.081	1.55e-04 ***

Problem 8



Problem 9



Problem 10

Medicaid expansion leads to a decrease in uncompensated care for hospitals in states that have adopted the change in policy. The event study graph is concerning because the bars for pre-treatment inch away from the zero margin, showing a violation of the parallel trends assumption. One primary limitation of the TWFE estimator is its biased nature in the context of staggered Medicaid expansion, as it doesn't take into account uncha